Bryce Grant | EE PhD Candidate

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EDUCATION & HONORS

Case Western Reserve University Cleveland, OH
PhD Student Robotics & CV Expected: May 2029
Graduate Advisor: Dr. Peng (Edward) Wang
Graduate Honors: *NSF Graduate Research Fellow (GRFP) Recipient, Advanced to Candidacy, NVIDIA Academic Grant*
University of Kentucky – *Magna Cum Laude, Lexmark Scholar* Lexington, KY
Dual B.S. in Computer Engineering and Electrical Engineering (CS, Math Minors) Graduated: May 2024

PUBLICATIONS & PREPRINTS

TrianguLang: Geometry-Aware Semantic Consensus for Pose-Free 3D Localization
B. Grant, A. Rothenburg, A. Banerjee, P. Wang.
Under Review | Website

Not All Features Are Created Equal: A Mechanistic Study of Vision-Language-Action Models
B. Grant, X. Zhao and P. Wang
Under Review | ICLR MM Intelligence Workshop, 2026. Website

Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representation
B. Grant and P. Wang
ICLR UCLR Workshop, 2026. Website

Quaternion Approximate Networks for Enhanced Image Classification and Oriented Object Detection
B. Grant and P. Wang
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. Website

EXPERIENCE

Mercor Remote
Applied AI Engineer Internship Aug. 2025 – Present

- Built a video QA pipeline for egocentric manipulation datasets, automating multi-level validation to ensure data readiness for sim-to-real transfer
- Built an internal analytics platform used by project leads to track individual contractor productivity, task accuracy, and budget allocation across multi-million dollar RLHF and SFT data collection operations
- Developed multimodal evaluation benchmarks across 30+ domains with difficulty gating and multi-condition evaluation, targeting tasks where frontier models had reached saturation
- Conducted failure analysis across frontier model evaluation trajectories, developing a structured taxonomy of failure modes that informed post-training data strategy and benchmark rebalancing

Case Western Reserve University Cleveland, OH
Graduate Research Assistant Aug. 2024 – Present

- Sequential Planning via Anchored Robotic Keypoints (SPARK): Achieved 90% & 36% zero-shot success on LIBERO & LIBERO PRO without training, outperforming code-generation &VLA baselines by 2x on out-of-distribution tasks
- Awarded an NVIDIA Academic Grant to pursue SPARK
- Developing a cooperative loco-manipulation framework for heterogeneous bimanual manipulation on UR10e and Unitree Go2, with sim-to-real transfer via learned world models
- Leading a group of undergraduate researchers on robotic perception and manipulation pipelines, guiding project scoping, code reviews, and experiment design

HP Vancouver, WA
PHD ML Intern May 2024 – Aug. 2024

- Deployed an anomaly detection pipeline on AWS for enterprise cloud spend analysis, combining LSTM forecasting with STL decomposition to flag cost anomalies across accounts
- Engineered a comparative RAG framework: benchmarked semantic vs. recursive chunking, BM25 vs. FAISS retrieval, and LLM query expansion strategies, with evaluation via coherence and NDCG metrics
- Built a customer-facing LLM chat agent that guided users through printer setup, replacing static manuals with dynamic reranking and multi-stage error recovery

University of Kentucky Lexington, KY
AI for Smart Manufacturing Lab Research Assistant Aug. 2023 – May. 2024

- Developed CNN-Transformer hybrid for 6-DOF pose estimation from RGB-D inputs for robotic arm manipulation
- Led navigation subsystem for autonomous telepresence robot, implementing 3D SLAM with servo-mounted LiDAR, achieving real-time mapping and dynamic obstacle avoidance in dynamic environments

Honeywell Atlanta, GA
Embedded Software Engineer Intern May. 2023 – Aug. 2023
- Developed bare-metal firmware API for life safety microcontrollers, supporting product line with 100K+ unit volume
- Validated dual-CPU communication protocols using oscilloscope debugging and hardware-in-the-loop testing

SKILLS

Programming Languages: C/C++, CUDA, MATLAB, Python, SQL
ML & Robotics: COLMAP, Distributed Training, HuggingFace, Isaac Lab, JAX, LiDAR, MuJoCo, OpenCV, Open3D, Pytorch, RL, ROS2, WandB, VLA Models
Cloud, DevOps & Engineering: Blender, Docker, Linux, PySpark, Snowflake
Domains & Methodologies: Benchmark Engineering, Causal Inference, Computer Vision, Geometric Deep Learning, LLM Evaluation, Mechanistic Interpretability, Robotics
Robotic Platforms: Unitree Go2, UR10e, UR5e

SERVICE

Conference Reviewer December 2024 – Present
- ECCV 2026, ICLR 2026, ICML 2026, IROS 2025, NAMRC 25-26

Mathworks Student Ambassador October 2024 – Dec. 2025
- Hosted Hackathons and educational events while increasing MATLAB brand awareness across campus

NSBE Lexington, KY
Region III Finance Chair May 2023 – May. 2024
- Orchestrated a regional career fair with 50+ participating companies while raising over $250,000 in revenue in six months